

“AI’s Coming! Are We Students Human-Ready?”– A Multi-stakeholder Study.

1. Introduction and Key Research Questions

As we look to ChatGPT and other AI tools, and every answer is just an Alexa call away, I always ask my mother about “Why do we need to memorize and study for tests? If machines do all the mathematical and scientific calculations with better speed and accuracy, what skills can us humans bring to the table?”, to which my mother said, “I don’t know!”. Traditional learning is dependent on rote learning, memory, speed and accuracy – skills in which humans cannot possibly compete with machines. What makes us truly human is our creativity, critical thinking, imagination, communication with other humans, our ability to emote, empathize and understand behaviors. Are we developing these skills in schools for the next generation? This study presents a statistical analysis of perceived soft skill development across three stakeholder groups: students, teachers, and workforce professionals.

Key Statistical question: Is there a gap between students, teachers, and workforce professionals in how they perceive readiness in key soft skills for an AI driven workforce?

Other research questions: Priority Skills: What soft skills are most valued by each group (students, teachers, workforce) for an AI-driven future? Impact of Extra-curriculars: Is there a connection between how confident students feel in their soft skills and the extracurricular activities they are involved in? Barriers to Teaching Soft Skills: What barriers exist in teaching soft skills (from a teacher perspective)?

2. Data Collection Methodology

To gather relevant data efficiently across the multiple participant groups, an online survey was administered via emails and text messages using **SurveyMonkey**.

2.1 Participant selection and sample design: The study sample was stratified into distinct subgroups based on participant type (students, teachers, and workforce professionals). To ensure balanced representation for statistical analysis, quota sampling was applied, aiming to include at least 30 participants from each group. However, the selection within each subgroup was not purely random. Instead, surveys were sent to students, teachers, and workforce professionals known to the researcher or the researcher's family. Teachers, on the other hand, were primarily recruited from the researcher's school. Participation was voluntary, and all responses were kept anonymous to promote honest and unbiased feedback. 37 workforce professionals and 31 students completed the survey, while only 9 teachers completed the entire survey.

2.2 Survey design: The survey instrument was carefully developed to measure perceptions across several key areas of soft skill development. The questionnaire included a mix of *Likert-scale questions* where students were asked to rate their confidence in several soft skills, teachers and workforce professionals were asked to assess the same soft skills of students and competencies of new hires respectively; and *Multiple-choice questions* to gather information on most valued/important soft skill, overall readiness and specific experiences by the three study groups. The soft skills focused on in the survey included communication, teamwork, adaptability, emotional intelligence, initiative and critical thinking.

3. Data Analysis: Data analyses of the multi-stakeholder survey were conducted using tools like Microsoft Excel and Python. Survey data was checked for missing values and data quality was verified. Descriptive statistics (Table 1) and box plot charts (Chart 1) shown below helped understand overall trends in how students, teachers, and workforce professionals rated soft skills.

Independent two-sample t-tests were conducted to compare differences between groups.

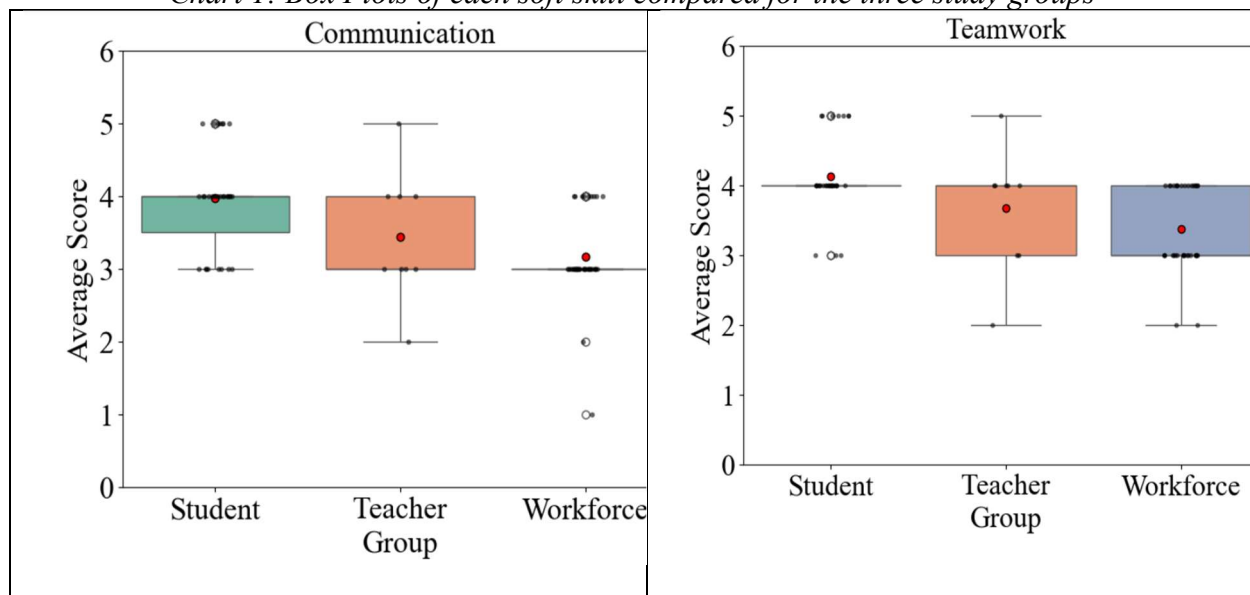
Correlation analysis was performed to see if participation in extracurriculars were linked to stronger soft skills.

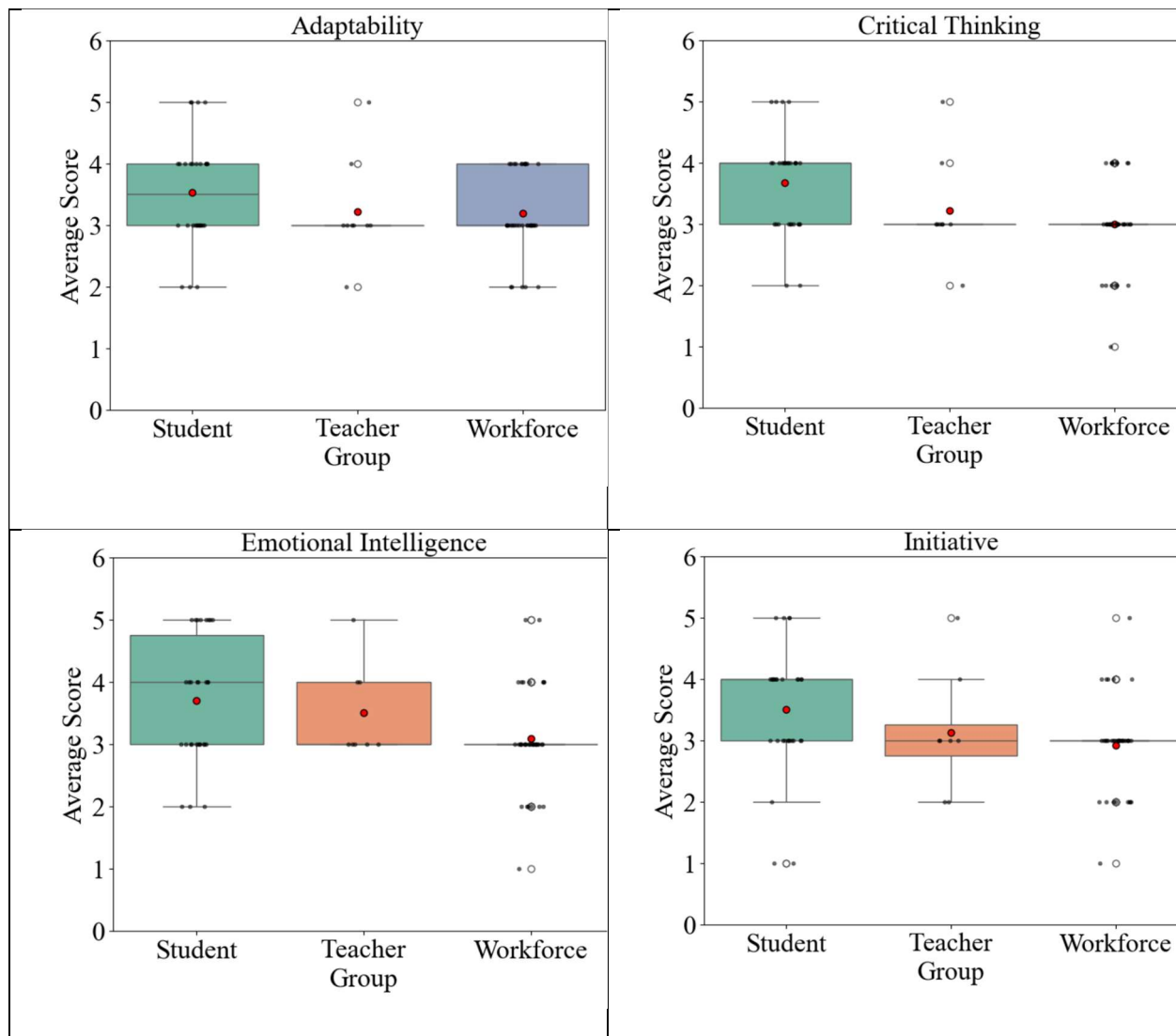
4.1 Data Display: Descriptive Statistics

Table 1: Mean Soft Skill Ratings by Group

Skill	Student Ratings of Confidence					Teacher Ratings of Students' skills					Workforce Ratings of New Hire skills				
	n	Mean	SD	95% CI Low	95% CI High	n	Mean	SD	95% CI Low	95% CI High	n	Mean	SD	95% CI Low	95% CI High
Communication	31	3.97	0.71	3.72	4.22	9	3.44	0.88	2.76	4.12	37	3.16	0.6	2.97	3.35
Teamwork	31	4.13	0.56	3.93	4.33	9	3.67	0.87	3	4.34	37	3.38	0.59	3.19	3.57
Adaptability	30	3.53	0.86	3.22	3.84	9	3.22	0.83	2.58	3.86	37	3.19	0.66	2.98	3.4
Critical Thinking	30	3.67	0.8	3.38	3.96	9	3.22	0.83	2.58	3.86	37	3	0.71	2.77	3.23
Emotional Intelligence	30	3.7	0.99	3.35	4.05	8	3.5	0.76	2.86	4.14	37	3.08	0.83	2.81	3.35
Initiative	30	3.5	1.01	3.14	3.86	8	3.12	0.99	2.29	3.95	37	2.92	0.76	2.68	3.16

Chart 1: Box Plots of each soft skill compared for the three study groups





4.2 t-test to study differences between students' readiness and workforce perceptions:

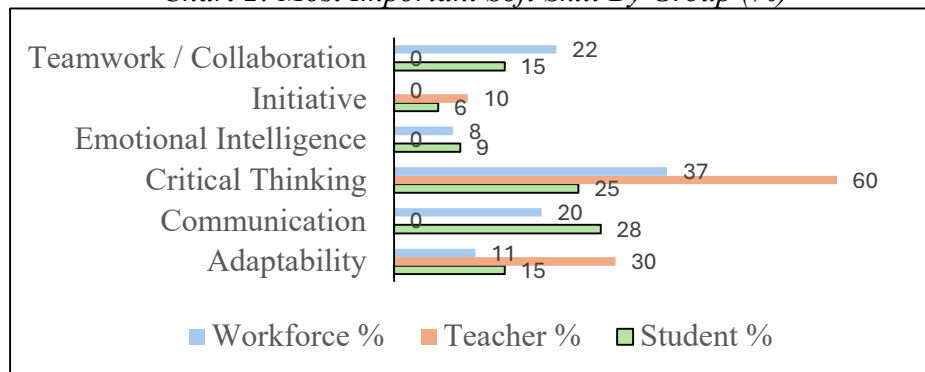
A two-sample independent t-test was conducted to compare soft skill ratings between students and workforce professionals. Teachers are not included in this analysis as we do not have sufficient sample size to make meaningful inferences. For most skills, there was a significant difference in means between students' and workforce ratings at the 95% confidence level, indicating a gap in perception between these groups. The results of the t-test are shown below in Table 2.

Table 2: Mean Soft Skill Ratings by Group

Skill	Group 1 (Students)	Group 2 (Workforce)	Difference of Mean	t- valu e	df	p- value	95% CI Diff	Significa nt?
Communication	3.97 ± 0.71	3.16 ± 0.60	0.81	5.01	59	0	[0.48, 1.13]	Yes
Teamwork	4.13 ± 0.56	3.38 ± 0.59	0.75	5.34	64	0	[0.47, 1.03]	Yes
Adaptability	3.53 ± 0.86	3.19 ± 0.66	0.34	1.8	53	0.0771	[-0.04, 0.73]	No
Critical Thinking	3.67 ± 0.80	3.00 ± 0.71	0.67	3.57	58	0.0007	[0.29, 1.04]	Yes
Emotional Intelligence	3.70 ± 0.99	3.08 ± 0.83	0.62	2.74	56	0.0083	[0.17, 1.07]	Yes
Initiative	3.50 ± 1.01	2.92 ± 0.76	0.58	2.61	52	0.0117	[0.13, 1.03]	Yes

4.3 Most important/valued skill: Chart 2 shows the percentages of each group and shows what the most important soft skill is from each group's perspective. Both teachers and workforce professional value critical thinking as the most important soft skill to be successful, whereas students rated Communication as the most important skill.

Chart 2: Most Important Soft Skill By Group (%)



4.4 Correlation Analysis: Correlation analyses was done using Pearson's's coefficient to examine the relationships between soft skills and extracurriculars and seeing which soft skills and extracurricular combinations have the strongest relationships. Being involved in Arts had a moderate to strong relationship with adaptability, and so did participation in Debate with

initiative. Participation in STEM/Coding showed a moderate correlation with critical thinking and adaptability. Chart 3 shows the correlation coefficients indicating relationships of all the activities and soft skills.

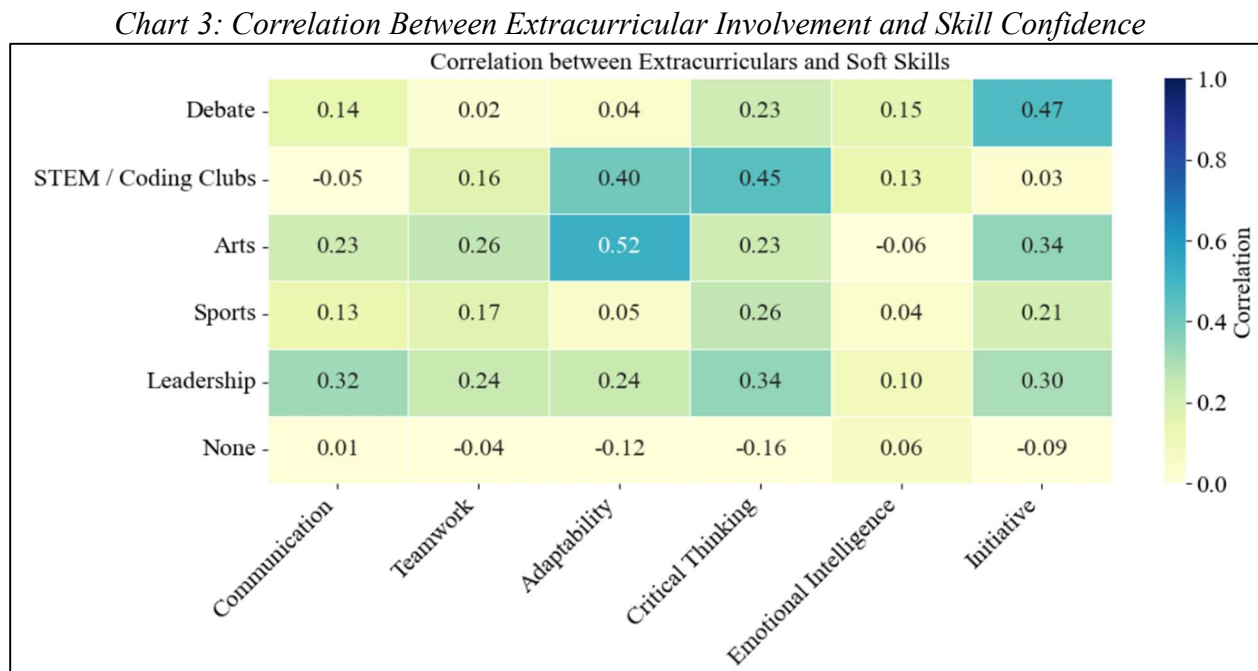


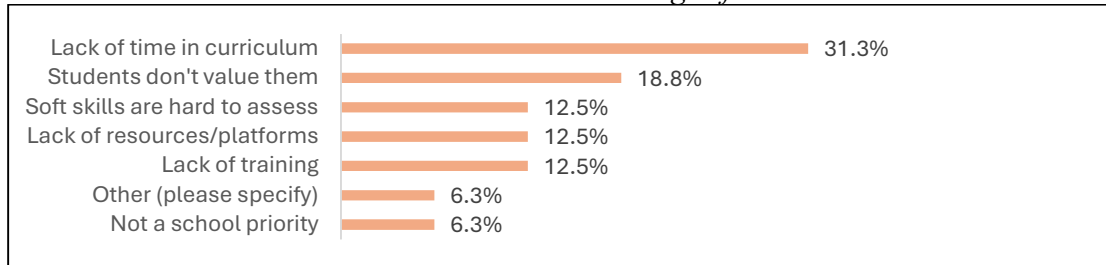
Table 3 shows the means of soft skill ratings and the students that do different extracurricular activities. The activity with the highest mean score in each soft skill is highlighted in green below.

Table 3: Mean Soft Skill Ratings v/s Extra-Curricular Activities

	Communication	Teamwork	Adaptability	Critical Thinking	Emotional Intelligence	Initiative
STEM/Coding Clubs	3.92	4.23	3.92	4.08	3.85	3.54
Leadership	4.2	4.27	3.73	3.93	3.8	3.8
Sports	4.08	4.25	3.58	3.92	3.75	3.75
Debate	4.07	4.14	3.57	3.86	3.86	4
Arts	4.14	4.29	4.00	3.86	3.64	3.86
None	4.00	4.00	3.00	3.00	4.00	3.00

Chart 4 shows the barriers and percentages, explaining why some teachers may find it difficult to teach soft skills in the classroom. As per teachers, lack of time in curriculum and students not valuing them were the biggest barriers in teaching soft skills in the classroom.

Chart 4: Barriers in teaching soft skills



5. Key Findings:

- Consistently, students rated themselves higher in soft skills than teachers and workforce professionals, thus showing greater confidence in their soft skills. Teachers' ratings of soft skills of students fall in between the self-report of students and the workforce professionals' perceptions of new hires.
- While students perceived communication as the most important skills required for the workforce, teachers and workforce professionals rated critical thinking as the most important soft skill for an AI driven workforce.
- Participating in extracurricular activities shows a moderate correlation with observed skill development, emphasizing its significance.

6. Conclusion: The data reveals a gap in the perceived development of soft skills, particularly between students and employers, indicating a disconnect between academic preparation and workplace expectations. Students often value different soft skills than the ones identified as essential by educators and employers. Additionally, these findings support the idea that extracurricular activities are valuable platforms for fostering essential soft skills among students.

Overall, this shows that schools should not only focus on academics but also help students build real-world skills through activities and better preparation for what the workforce expects.

Recommendations: To bridge the soft skill gap, educators should be recommended to use more project-based, collaborative learning, while workforce professionals can partner with schools to provide real-world training and mentorship. Students are encouraged to take initiative outside the classroom to actively develop skills like communication, teamwork, and adaptability.

Reflections on Process and Limitations: While this project gave some interesting results, there were some limitations. A large portion of the data came from students' self-ratings, and not a true assessment of their skills. Students tend to overestimate their abilities, as compared to what the workforce expects. In the future, it would be helpful to include actual tests to gauge skill development and to get a more practical assessment of soft skills. Furthermore, the number of people who participated in the survey was very limited and were mostly from the same school or region. If more students, teachers and professionals from different areas and backgrounds were included, the results would have been much more diverse, resulting in better representation and reliability.

Future Research Suggestions: To build on this project, future research could include longitudinal tracking of soft skill development, adding qualitative interviews or open-ended questions to supplement quantitative data that would also help explain the "why" behind the survey results, giving more depth to the data.

8. References:

1. Ramnanan, Naveen, "Developing Soft Skills: Faculty and Employer Perspectives and Recommendations" (2022). *Walden Dissertations and Doctoral Studies*. 12829.

2. Dean, Susan & East, Julia. (2019). Soft Skills Needed for the 21st-Century Workforce.

International Journal of Applied Management and Technology. 18.

10.5590/IJAMT.2019.18.1.02.

Parental guidance: Researcher took help from parents in selection of participants from the workforce and distributing surveys to them, guidance with developing box plots in Python, and overall review of statistical tools and methodology.

9. Appendix: Raw Data for Key Questions

1. What is the most important soft skill to develop as we gear up for an AI-driven future?

	Students	Teachers	Workforce
Communication	9	0	7
Teamwork /Collaboration	5	0	8
Adaptability	6	3	4
Initiative	2	1	3
Critical Thinking	8	6	13
Emotional Intelligence	3	0	3
Missing	5	0	0
Total	38	10	38

2. On a scale of 1 to 5, how confident are you in [your /new hires'/students'] ability to:?

Student Rating	Extremely dissatisfied	Dissatisfied	Somewhat satisfied	Very satisfied	Extremely satisfied	Missing
Communication	0	0	8	16	7	7
Teamwork	0	0	3	21	7	7
Adaptability	0	3	12	11	4	8
Critical Thinking	0	2	10	14	4	8
Emotional Intelligence	0	3	11	8	8	8
Initiative	2	1	11	12	4	8

Teacher Rating	Extremely dissatisfied	Dissatisfied	Somewhat satisfied	Very satisfied	Extremely satisfied	Missing
Communication	0	1	4	3	1	1
Teamwork	0	1	2	5	1	1

Adaptability	0	1	6	1	1	1
Critical Thinking	0	1	6	1	1	1
Emotional Intelligence	0	0	5	2	1	2
Initiative	0	2	4	1	1	2

Workforce Rating	Extremely dissatisfied	Dissatisfied	Somewhat satisfied	Very satisfied	Extremely satisfied	Missing
Communication	1	1	26	9	0	1
Teamwork	0	2	19	16	0	1
Adaptability	0	5	20	12	0	1
Critical Thinking	1	6	22	8	0	1
Emotional Intelligence	1	6	21	7	2	1
Initiative	1	8	22	5	1	1

3. What type of extra-curricular activities do you participate in? [Select all that apply]

	Students
Debate	14
STEM / Coding Clubs	13
Creative arts (Visual arts, music, theater)	14
Sports & Athletics	12
Leadership roles	15
None of the above	1

4. What barriers exist in teaching soft skills? [Select all that apply]

	Teachers
Lack of time in curriculum	5
Lack of training	2
Lack of resources/platforms	2
Soft skills are hard to assess	2
Students don't value them	3
Not a school priority	1
Other (please specify)	1